

Geometry-Aided Kalman-Based Tracking Loops for Ionospheric Scintillation Monitoring

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Outline

1 Introduction

2 Introduction

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Ionospheric Scintillation

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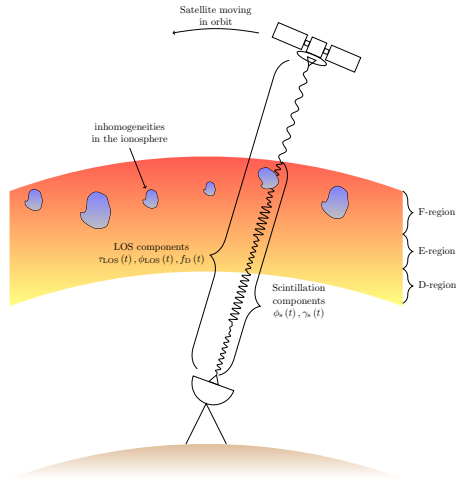


Figure: Ionospheric scintillation – an overview.

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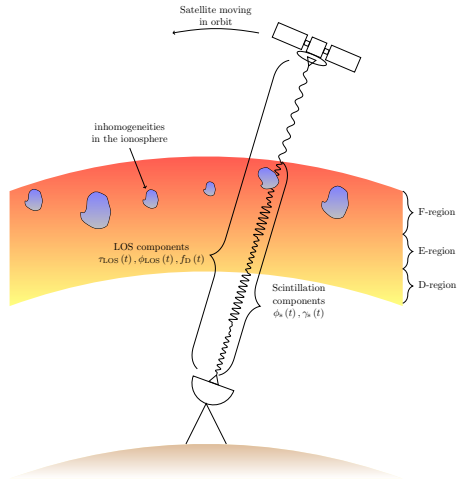


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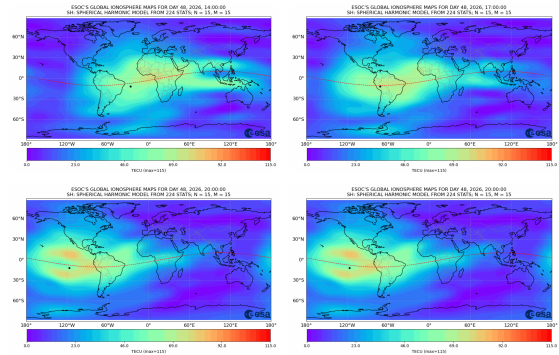
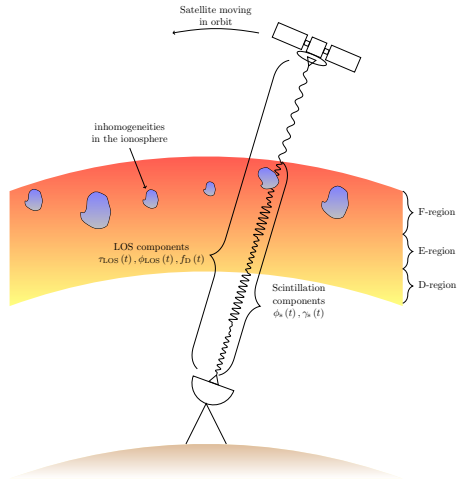


Figure: Temporal evolution of the TEC unit index.

Figure: Ionospheric scintillation – an overview.

Introduction

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$$R = \left(\frac{A}{aL} \right)^{\frac{1}{b}} \quad (1)$$

Applying the $\ln$ transformation in both side of this equation, we have

$$\ln R = \frac{1}{b} \ln A - \frac{\ln(aL)}{b} \underbrace{\ln R}_y = \underbrace{\frac{1}{b}}_{\theta_1} \underbrace{\ln A}_x - \underbrace{\frac{\ln(aL)}{b}}_{\theta_0}$$

Hence, we have a linear model, i.e., $y = \theta_1 x + \theta_0$.

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teste

visible Final test

But uncover both occupies space and is

teste

Only: Neither occupies space nor is visible

But uncover both occupies space and is visible Final test

teste

Visible: Occupies space but is invisible But uncover both occupies space and is visible Final test

teste

visible Final test

But uncover both occupies space and is

Master Dissertation highlight's

Objetivo geral: Concepção da arquitetura digital do modem AFSK. Implementação do protótipo lógico no *software MATLAB/Simulink*.

Objetivos específicos:

- Descrição matemática de uma nova arquitetura de modulador AFSK.

- Derivação do limite superior para a probabilidade de erro.

- Estimadores de fase e tempo para a nova arquitetura.

Undergraduated in Electronic Engineering at Unifor(Universidade de Fortaleza)	2013-2018
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teste

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Who am I?

Some personal information:

Rubem Vasconcelos Pacelli, 27 years old.

I am from Fortaleza, Ceará, Brazil.

Academic information:

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Jan 2013 – Dec 2018

MSc in Teleinformatics Engineering
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Programming (and hardware description) Languages:

Matlab/Simulink: 5 years of experience, approximately.

Python: 2 years of intense usage.

C, C++, VHDL, and Java: Secondary usage.

Some further technical skills:

Git: Elementary concepts.

$\text{\LaTeX}$: Elementary concepts.

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Main challenges

The design of modern digital modems for the further development of CubeSat satellite systems. The Viterbi algorithm is employed for the maximum-likelihood sequence detection, and its stages are software-defined (SDR) concept, combined with the enhancement of digital signal processing techniques, FPGA (field-programmable gate array), and microelectronics, has motivated a new generation of TT&C (telemetry, tracking and command) modules that brings hardware flexibility, but at the same time, matches with the current modulation schemes.